This way, we partition the plane in rectangular areas. With respect to our initial idea of splitting points with vertical lines only, we have fewer areas extending to infinity (while if we always used the same coordinate all areas would be infinite!) and avoid keeping distant points in the same partition.

At the same time, each node has just 2 children, and we can maintain all the advantages and guarantees of binary partitioning and binary search.

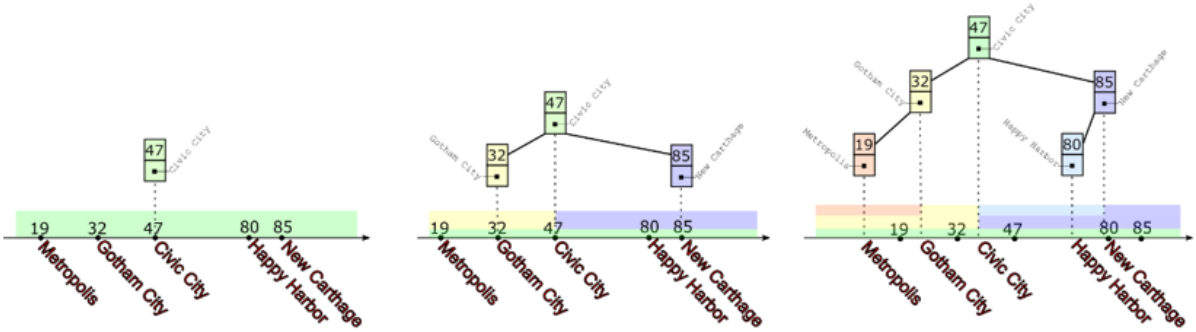


Figure 9.3 Constructing a BST from the pivots of a 1-D dataset.
(Left) We add the first pivot, that is going to be the root of the tree. The pivot creates an implicit division along the x axis, partitioning the remaining points into left and right subsets. The region covered by a node is the union of the partitions its pivot creates, so the root covers the whole subset.
(Center) Each of the two sub-regions implied by the root is further partitioned, by selecting a point in each region as pivot. As shown by the highlighting above the horizontal axis, now each of these nodes at level 1 covers half of the space (while the root still covers the whole space).
(Right) Level 2 nodes are added, further partitioning